

Technical Datasheet of WIN-IO-16UI8UOM Module (Modbus RTU)

1. Overview

The Universal Input (UI) and Universal Output (UO) board support **communication exclusively over Modbus RTU**, enabling seamless control and monitoring of inputs/outputs. The board provide **configurable input and output selection via RTU**, ensuring flexibility across various applications.



The module acts as an input/output device between the Modbus RTU Serial. Digital IO module is study, low power usage and easy to use.

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

Specifications

<u>Category</u>	<u>UIUO Module</u>
Input Type	Analog Input (10V) Analog Input (5V) Digital Input NTC Sensor Input. (Configurable via RS485)
Output Types	Analog Output (AO), Digital Output (DO). (configurable via RS485)
Connectors	Single port handling multiple input/output types

General –

Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 9 – 24 V DC or 24 V AC/DC Typical – 24V DC @ 700mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



UI (Universal Inputs) –

Digital Input channels	16
Sense Voltage	3.3 – 12 VDC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec
Analog Input channels	8
Input Signal	0 – 5VDC / 0 – 10VDC / NTC (Web page selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 15 VDC min
AI input impedance	120Ω

UO (Universal Outputs) –

Digital Output channels	8
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55°C~+150°C

Analog Output channels	8
Analog Resolution	10 bits (0 to 10VDC)
Accuracy	± 2% Full scale
Isolation	Optically Isolated
Conversion Time	15 msec

Additional Features: -

Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air

Configuration Of UIUO module:-

The UI/UO module supports flexible I/O configuration. Each of the 16 UI (Universal Input) UI1 to UI8 ports can be independently configured as Analog Input, NTC Sensor Input, Digital Input and UI9 to UI16 must be configure as Digital Input. Similarly, each of the 8 UO (Universal Output) ports can be independently configured as, Analog Output, Digital Output.

All the configuration can be done through the RS485 as Mbpoll.

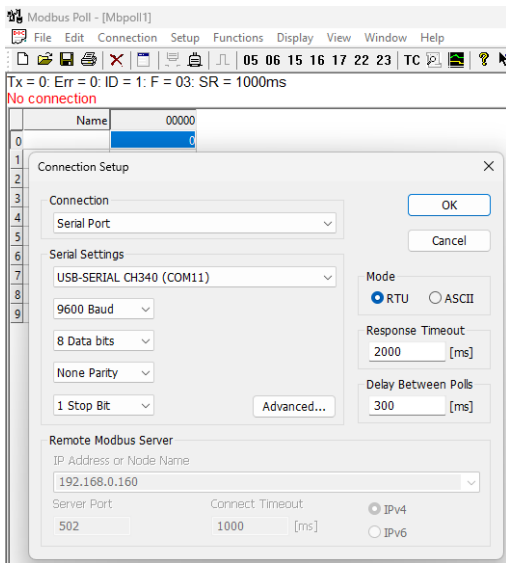


Image1

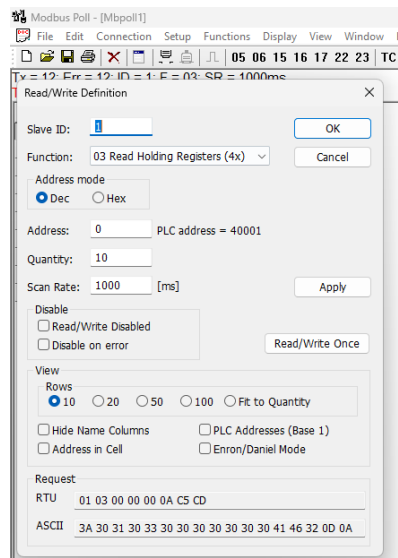


Image2

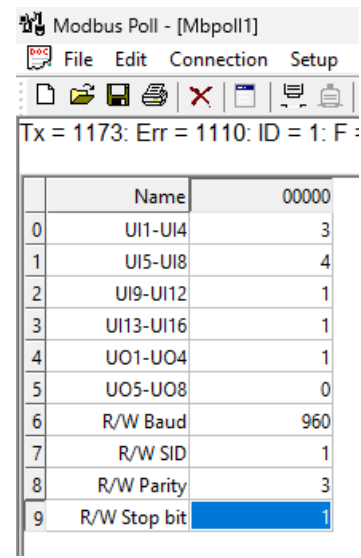


Image3

The following are step by step guide for configuration:

- Connect the mbpoll with default addresses as Baud-9600, SID-1, Parity-None, Data bit-8 as shown in image 1 & 2.
- Click on setup, then Read/Write definition there func code should be 03.
- Address 0 and quantity 10 for setup mode.
- 40001 to 40006 registers for setup UI/UO modes.
- 40007 to 40010 registers for soft setting SID/Baud/Parity/Data bit.

Setup mode of UIUO module:-

- Refer Above Image 3 for Setup mode
- For Universal Input enter 1 as DI, 2 as 10v, 3 as NTC, 4 as 5v
- For UI setup Fun code 03 and Modbus register from 40001 to 40004
- At 40001 reg UI1 to UI4 channels will be set, enter 1/2/3/4 as required in reg 40001
- At 40002 reg UI5 to UI8 channels will be set, enter 1/2/3/4 as required in reg 40002
- At 40003 reg UI9 to UI12 channels will be set, must enter 1 in reg 40003
- At 40004 reg UI13 to UI16 channels will be set, must enter 1 in reg 40004
- For example, to set Channel 1 to 4 of UI as NTC Sensor input then in reg 40001 enter 3.

Note: Always press reset button in the device after doing configuration for UI/UO Status, then only changes in configuration will take place.

- For Universal Output enter 0 as AO, 1 as DO
- For UO setup Fun code 03 and Modbus register from 40005 to 40006
- At 40005 reg UO1 to UO4 channels will be set, enter 1/2 as required in reg 40005
- At 40006 reg UO5 to UO8 channels will be set, enter 1/2 as required in reg 40006
- For example, to set Channel 1 to 4 of UO as Digital output (DO) then in reg 40005 enter 1.

Note: Always press reset button in the device after doing configuration for UI/UO Status, then only changes in configuration will take place.

Read/Write mode of UIUO module:-

1. For Universal Input

- For Universal Input Read Mode Fun code 03 and Modbus registers from 40011 to 40026.
- For DI the data will be 1/0 at selected registers.
- For AI data will be 0-1023 resolution at selected registers.
- For NTC data will be temperature ex-27.6 as 276 at selected registers.

Note: The data of UI port can be fetched from the mentioned Modbus registers, irrespective of DI, AI, NTC, selection. The Desired Inputs are as follows:

INPUT TYPE	Input Value
DI	0 V(0) , 12VDC (1)
AI(5V)	0 to 5VDC (0-1023)
AI(10V)	0 to 10VDC (0-1023)
NTC	NTC sensor(Temperature in °C)

2. For Universal Output

- For Universal Output Write Mode Fun code 03 and Modbus registers from 40027 to 40034.
- For DO the data will be write 1/0 at selected registers.
- For AO the data will be write 0 to 1023 resolution at selected registers.

Note: Similarly, the data of the UO port can be set in the same register as mentioned Modbus register. For DO, It should be sent 0 or 1 (12VDC). And for AO a 10-bit resolution value must be sent (i.e. from 0 to 1023) to set output of 0 to 10VDC.

OUTPUT Type	Output Value
DO	0 (0V) , 1 (12VDC)
AO	0 to 10VDC (0-1023)

UIUO Status:-

Modbus Poll - [Mbpoll1]

File Edit Connection Setup

Tx = 1173: Err = 1110: ID = 1: F :

	Name	00000
0	UI1-UI4	3
1	UI5-UI8	4
2	UI9-UI12	1
3	UI13-UI16	1
4	UO1-UO4	1
5	UO5-UO8	0
6	R/W Baud	960
7	R/W SID	1
8	R/W Parity	3
9	R/W Stop bit	1

- Here UI1 to UI4 selected as NTC(3), UI5 to UI8 selected as AI(5v) and UI9 to UI16 selected as DI(1).
- And UO1 to Uo4 selected as DO(1) , UO5 to UO8 selected as AO(0).

Modbus Poll - [Mbpoll1]

File Edit Connection Setup Functions Display View Win

Tx = 211: Err = 0: ID = 1: F = 03: SR = 1000ms

	Name	00010	Name	00020
0	UI1	255	UI11	1
1	UI2	-33	UI12	0
2	UI3	102	UI13	0
3	UI4	103	UI14	0
4	UI5	0	UI15	0
5	UI6	0	UI16	0
6	UI7	397		
7	UI8	0		
8	UI9	0		
9	UI10	0		

Image1

Modbus Poll - [Mbpoll1]

File Edit Connection Setup Functions Display View Windo

Tx = 515: Err = 0: ID = 1: F = 03: SR = 1000ms

	Name	00020	Name	00030
0			UO5	1023
1			UO6	512
2			UO7	0
3			UO8	900
4				
5				
6	UO1	1		
7	UO2	1		
8	UO3	0		
9	UO4	1		

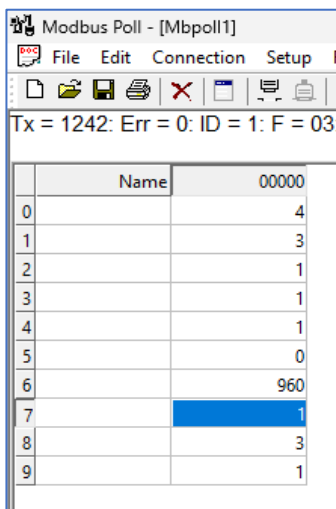
Image2

- In image1 shows the UI data as selected channels.
- At UI1 there is NTC sensor providing value as 255(25.5 °C)
- At UI7 there is AI sensor providing value as 397 resolution.
- At UI11 there is DI sensor providing value as 1(HIGH)
- In image2 shows the UO data as selected channels.
- At UO1 to UO4 there is DO provided value as 1/0.
- At UO5 to UO8 there is AO provided value as 0-1023.

Modbus Soft Setting Configuration:-

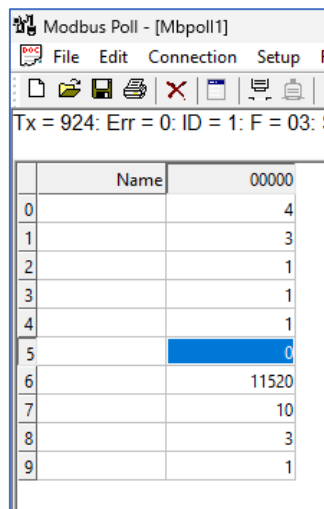
Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below image1.
- As you can see in the image1 below, the resistor 40007 – 960 represents the present baud rate (9600) of the board and the resistor 40008 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed as shown in image3 below, then you must enter a value on the 40007 register for the baud rate and on the 40008 register for the slave id as shown in the image2 below, in single register we can Read & Write the values.
- Now you can see data. In below image2 set 115200 baud rate and slave ID 10.



	Name	00000
0		4
1		3
2		1
3		1
4		1
5		0
6		960
7		1
8		3
9		1

Image1



	Name	00000
0		4
1		3
2		1
3		1
4		1
5		0
6		11520
7		10
8		3
9		1

Image2



Image3

- As you can see in the image1/2 above, the registers 40001 to 40004 shows the UI setup, registers 40005 to 40006 shows the UO setup and registers 40007 to 40010 represents the Modbus soft setting Baud/SID/Parity/Data bit.

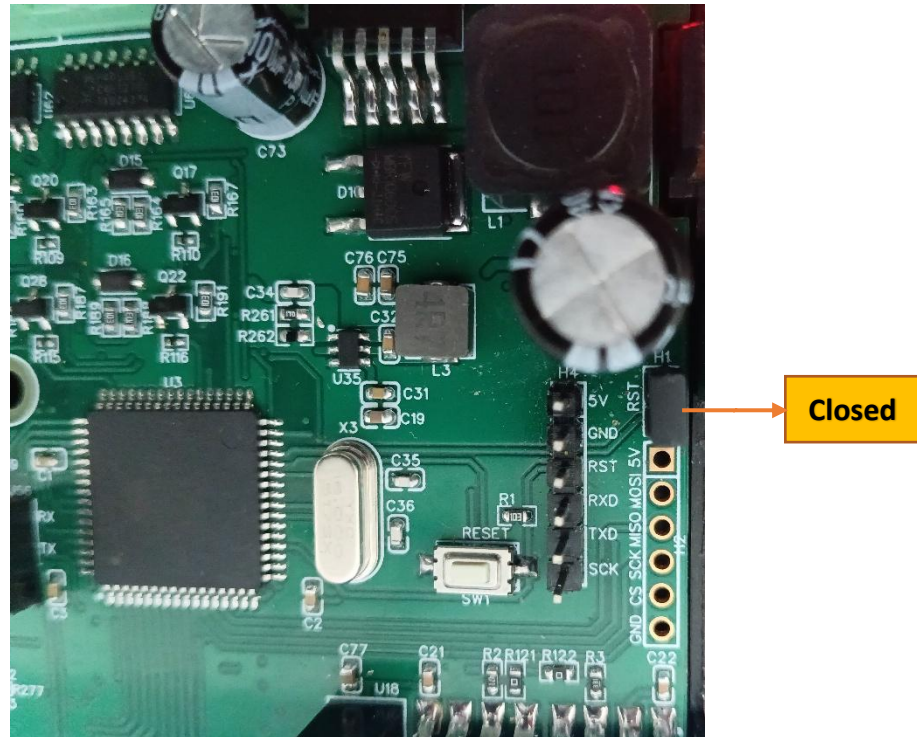


Image4

Hard reset Setting

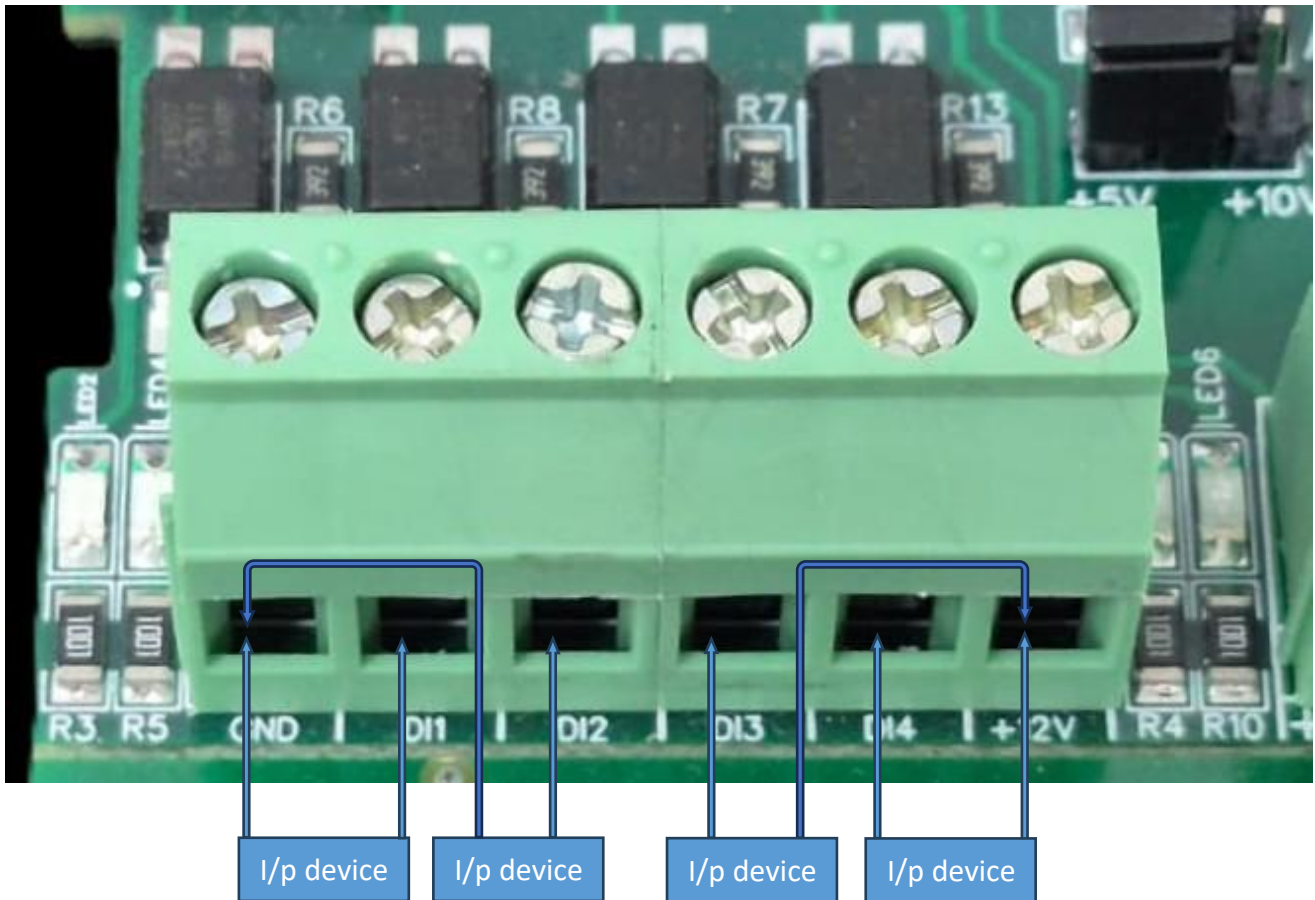
Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the above image4) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

MODBUS REGISTERS TABLE FOR UIUO PORTS:-

ID	Function Description	Register Description	Modbus Function Code	Data Type
1	SET UI1 TO UI4	40001	0x03	Boolean
2	SET UI5 TO UI8	40002	0x03	Boolean
3	SET UI9 TO UI12	40003	0x03	Boolean
4	SET UI13 TO UI16	40004	0x03	Boolean
5	SET UO1 TO UO4	40005	0x03	Boolean
6	SET UO5 TO UO8	40006	0x03	Boolean

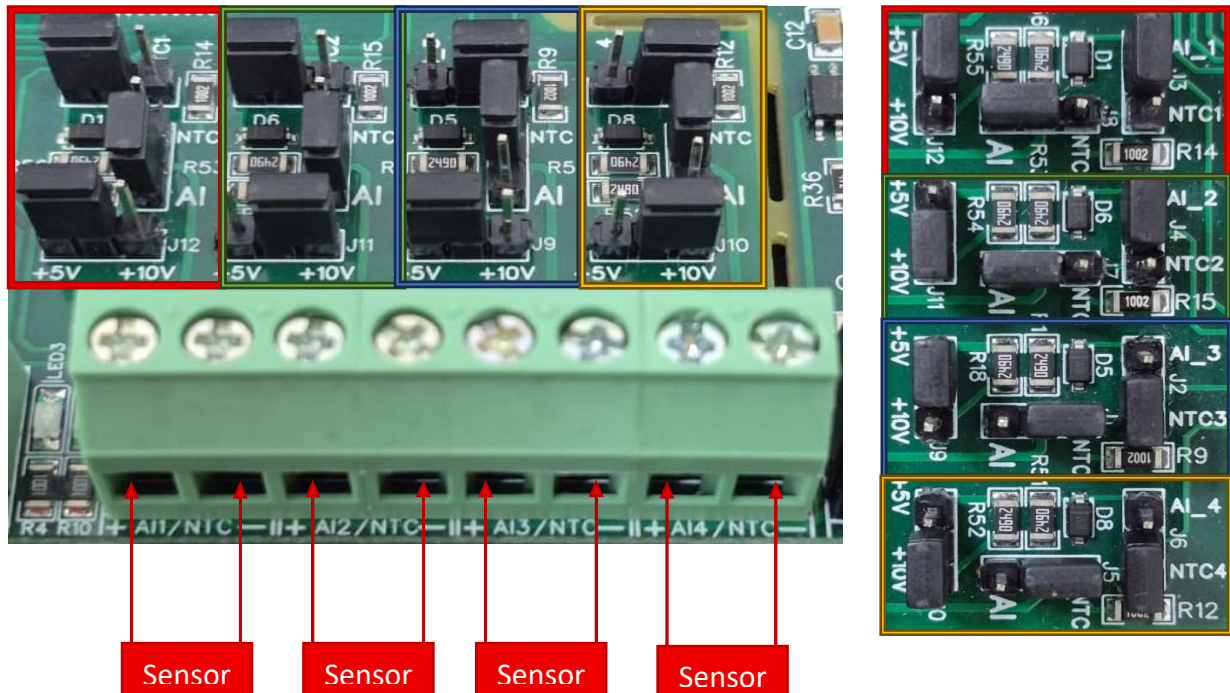
7	R/W BAUD	40007	0x03	Boolean
8	R/W SID	40008	0x03	Boolean
9	R/W PARITY	40009	0x03	Boolean
10	R/W STOP BIT	40010	0x03	Boolean
11	UI 1 (DI/AI/NTC)	40011	0x03	Boolean
12	UI 2 (DI/AI/NTC)	40012	0x03	Boolean
13	UI 3 (DI/AI/NTC)	40013	0x03	Boolean
14	UI 4 (DI/AI/NTC)	40014	0x03	Boolean
15	UI 5 (DI/AI/NTC)	40015	0x03	Boolean
16	UI 6 (DI/AI/NTC)	40016	0x03	Boolean
17	UI 7 (DI/AI/NTC)	40017	0x03	Boolean
18	UI 8 (DI/AI/NTC)	40018	0x03	Boolean
19	UI 9 (DI)	40019	0x03	Boolean
20	UI 10 (DI)	40020	0x03	Boolean
21	UI 11 (DI)	40021	0x03	Boolean
22	UI 12 (DI)	40022	0x03	Boolean
23	UI 13 (DI)	40023	0x03	Boolean
24	UI 14 (DI)	40024	0x03	Boolean
25	UI 15 (DI)	40025	0x03	Boolean
26	UI 16 (DI)	40026	0x03	Boolean
27	UO1(DO/AO)	40027	0x03	Boolean
28	UO2(DO/AO)	40028	0x03	Boolean
29	UO3(DO/AO)	40029	0x03	Boolean
30	UO4(DO/AO)	40030	0x03	Boolean
31	UO5(DO/AO)	40031	0x03	Boolean
32	UO6(DO/AO)	40032	0x03	Boolean
33	UO7(DO/AO)	40033	0x03	Boolean
34	UO8(DO/AO)	40034	0x03	Boolean

DI



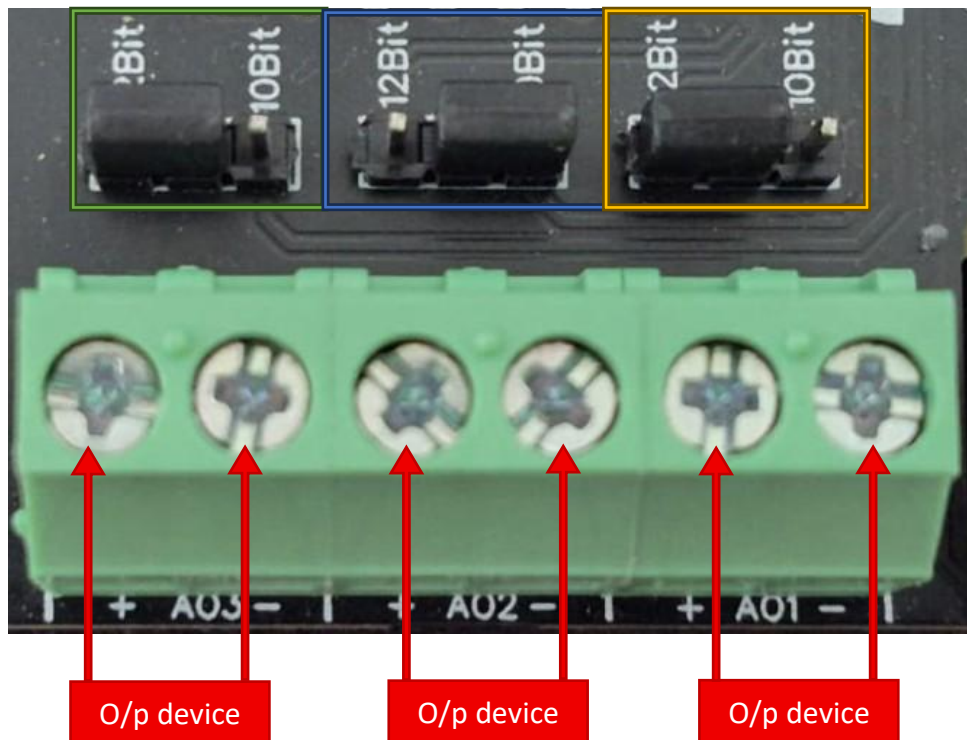
- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

AI-NTC



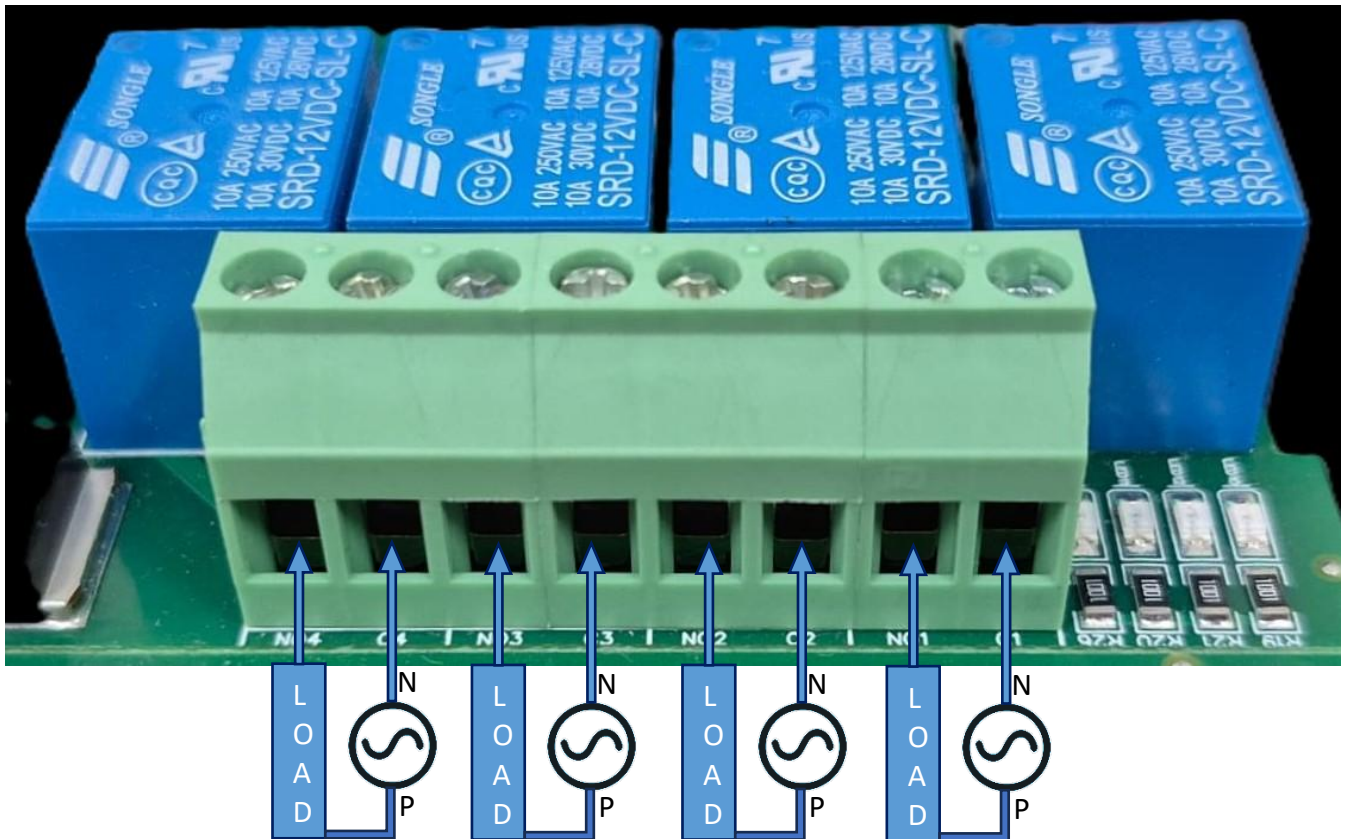
- **Analog Input & NTC Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to interface with either standard external analog sensors or NTC thermistors.
- **Channel Configuration (Visual Guide):** Each of the four channels features two dedicated jumpers to independently select the operating mode (AI or NTC) and the reference voltage range (5V or 10V). Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (AI Mode, 5V Range):** Set the mode jumper to **AI** and the voltage jumper to **+5V**.
 - **Green Box (AI Mode, 10V Range):** Set the mode jumper to **AI** and the voltage jumper to **+10V**.
 - **Blue Box (NTC Mode, 5V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+5V**.
 - **Yellow Box (NTC Mode, 10V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+10V**.
- **Sensor Wiring Connections:**
 - The sensor or thermistor's positive signal wire should be connected to the + terminal of the desired channel (labelled AI1/NTC, AI2/NTC, etc.).
 - The sensor's common or ground wire should be connected to the corresponding - terminal.
- **Crucial Setup Recommendation:** Always ensure that both the mode jumper (NTC vs. AI) and the voltage jumper strictly match the hardware you are connecting. Incorrectly setting an NTC to an AI mode, or mismatching the voltage limits, will result in inaccurate temperature or signal readings.

AO



- **Analog Output (AO) Operation:** Each channel provides two terminals: + (**Positive**) and - (**Negative/Common**). These are used to send variable analog signals to **external output (O/p) devices**, such as variable frequency drives or proportional valves.
- **Channel Configuration (Visual Guide):** Each of the three AO channels features a **dedicated jumper** to set the **output bit resolution**. Refer to the color-coded boxes in the board diagram to configure your desired resolution:
 - **Green & Yellow Box (12-bit):** Set the resolution jumper to **12Bit**.
 - **Blue Box (10-bit):** Set the resolution jumper to **10Bit**.
- **Output Device Wiring Connections:**
 - The external device's positive signal wire should be connected to the + terminal of the desired channel (AO1, AO2, or AO3).
 - The device's common or ground wire should be connected to the corresponding - terminal..
- **Crucial Setup Recommendation:** Always ensure that the hardware jumper setting (10-bit or 12-bit) matches the output scaling configured in your main control software. A mismatch between software resolution and hardware configuration will result in incorrect output signal levels.

DO

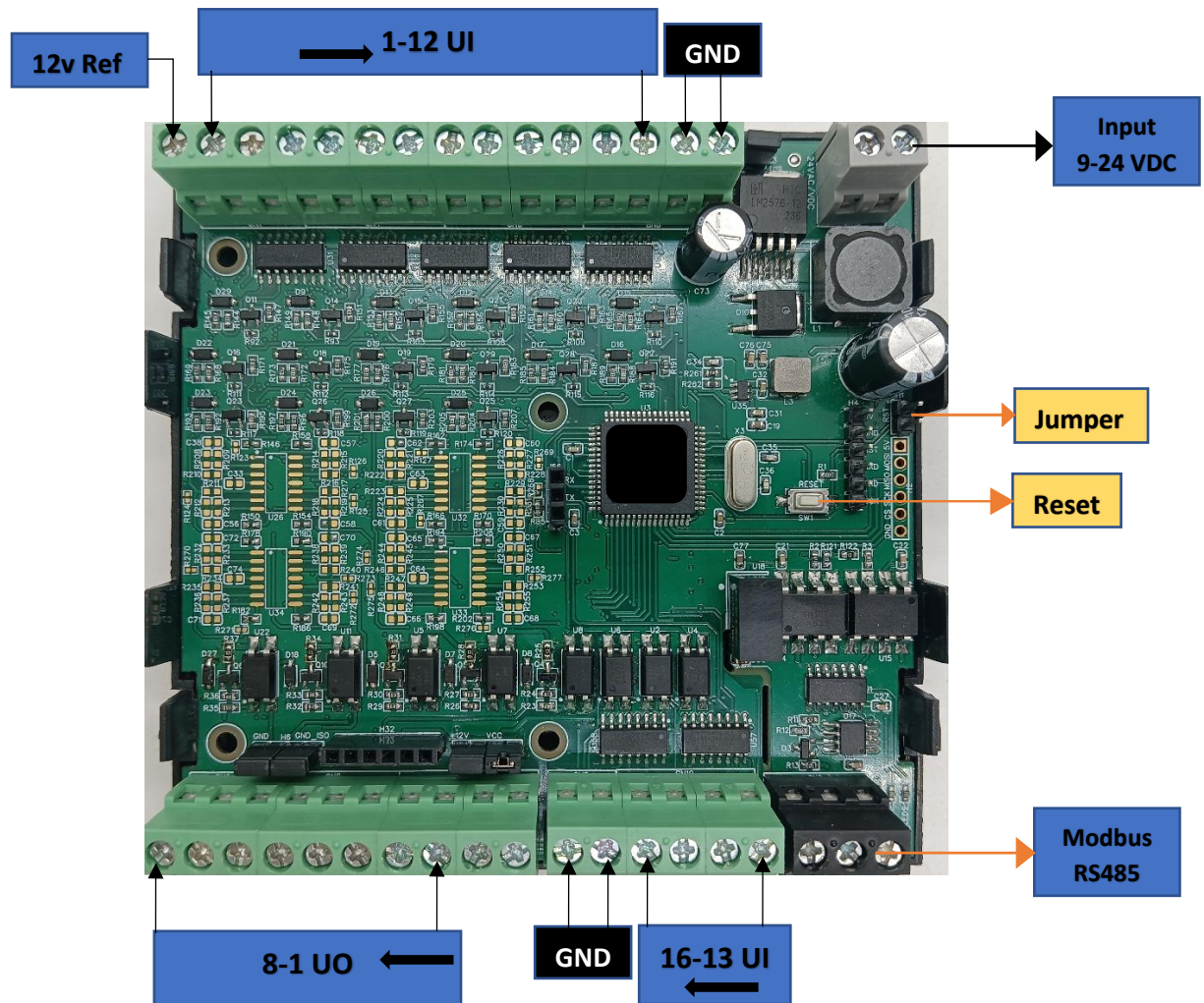


- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains **open**, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

Wiring Diagram:-

The Port number and Sequence of UI and UO are as mentioned in the wiring diagrams shown below. Also the position of default reset button is as shown in the diagram.



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